



Department of Computer Science and Engineering
Final Examination Fall 2024
CSE 321: Operating Systems

Duration: 2 Hours

Total Marks: 45

Answer the following questions.
 Figures in the right margin indicate marks.

1. CO1	Shoumo implemented an application consisting of 68% parallel portion and the application is designed in such a way that during the execution all of the threads created by it get accessed with only one kernel thread. I. If the number of cores of a system is increased from 7 to 9 then calculate the change of the performance of the application. II. Identify with proper explanation about the multithreading model which is being used by the application. Mention the drawbacks of the model.	[2+2]
2. CO2	a) A software company uses a centralized version control system to manage files collaboratively. To prevent conflicts, only one developer is allowed to modify a specific file at any given time. Before making changes, a developer checks if the "Edit Lock" is available. If the lock is available, the developer acquires it, indicating they are working on the file. This ensures that no other developer can modify the file simultaneously. Once the developer completes their changes, they release the "Edit Lock," allowing others to access the file for editing. I. Identify and explain which synchronization method has been used here? II. Logically detect which synchronization problems can be solved by this method?	[2+1]

	b) Find outputs of the given code: <pre> void *func1(int *n); void *func2(int *n); int shared=50; sem_t s1,s2; int main(){ pthread_t t[2]; sem_init(&s1,0,1); sem_init(&s2,0,0); pthread_create(&t[0],NULL,func1,NULL); pthread_create(&t[1],NULL,func2,NULL); for(int i=0;i<2;i++){ pthread_join(t[i],NULL); } sem_destroy(&s1); sem_destroy(&s2); printf("Final Value of Shared: %d\n",shared); return 0; } void *func1(int *n){ sem_wait(&s2); for(int i=0;i<2;i++){ printf("Current Value of Shared: %d\n",shared); shared=shared-5; } sem_post(&s2); } void *func2(int *n){ sem_wait(&s1); for(int i=0;i<3;i++){ printf("Current Value of Shared: %d\n",shared); shared=shared+3; } sem_post(&s1); sem_post(&s2); } </pre>	[3]
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[5]

c) Kabbya, an operating system developer, was experimenting with process synchronization. He came to know of Peterson's solution to the process synchronization problem. In the original solution, there was a variable named turn and an array named flag. He decided to use an array as the turn variable and implement his solution. The code and specifications of his problem is below:

- There are two processes: P0 and P1.
- P1 will start its execution first.
- Each assignment statement takes **4 ms** to execute.
- Each boolean logic statement takes **2 ms** to execute.
- Loop statement takes **1 ms** to execute.
- Context Switch will occur after **12 ms**.
- Critical section contains **2** assignments and **2** logical statements.
- Remainder section contains **2** assignment statements.
- For P0: i=1 and j=0.
- For P1: i=0 and j=1.
- turn[0]=FALSE, turn[1]=FALSE.
- flag[0] = FALSE, flag[1] = FALSE.
- If context switch occurs in the middle of executing a statement, the statement is abandoned and restarted again when the process gets the CPU again.

The code used to solve the problem:

```
do{
    flag[i] = true;
    turn[j] = true;
    while(flag[j] == true && turn[j] ==true);
    //critical section
    turn[i] = false;
    flag[i] = false;
    //remainder section
} while(true);
```

Complete the table below for processes P0 and P1 using **Peterson's solution**.

Process 0: i = 1, j = 0	Process 1: i = 0, j = 1
	

3. CO3	a) If we consider Dynamic Partitioning in memory, it will look like the following snapshot at a certain time where gray portions of the memory represent occupied spaces. Illustrate the processes in the memory with the space requirement of P1=100K, P2=300K, P3=200K, P4=145K, P5=150K, P6=60K and P7=50K (in order) applying worst fit and first fit algorithms. Calculate the external fragmentation if there is any. Explain which algorithm makes the most effective use of memory and justify your answer. <table data-bbox="424 526 1209 589"> <tr> <td>160K</td><td>250K</td><td>400K</td><td>50K</td><td>200K</td><td>260K</td></tr> </table>	160K	250K	400K	50K	200K	260K	[3+3 +1]																								
160K	250K	400K	50K	200K	260K																											
	b) Consider a system with a TLB hit ratio of 85%, and a memory access time of 150 ns. If Effective Access Time is 253 ns, then calculate the TLB access time.	[3]																														
4. CO3	a) In a system the size of the main memory is 64 Bytes and size of each page is 8 Bytes. In a certain moment the CPU generates logical addresses 21, 36, 3, 30, 41 and 15 of a process respectively. Logical memory and page table of the process is given below. <table data-bbox="336 1032 545 1473"> <tr><th colspan="2">Logical Memory</th></tr> <tr><td>0</td><td>110</td></tr> <tr><td>1</td><td>111</td></tr> <tr><td>2</td><td>220</td></tr> <tr><td>3</td><td>221</td></tr> <tr><td>4</td><td>230</td></tr> <tr><td>5</td><td>260</td></tr> </table> <table data-bbox="689 1032 896 1473"> <tr><th colspan="2">PMT</th></tr> <tr><td>0</td><td>6</td></tr> <tr><td>1</td><td>3</td></tr> <tr><td>2</td><td>0</td></tr> <tr><td>3</td><td>8</td></tr> <tr><td>4</td><td>7</td></tr> <tr><td>5</td><td>12</td></tr> </table> <table data-bbox="1038 1032 1246 1473"> <tr><th>Main Memory</th></tr> <tr><td style="height: 200px;"></td></tr> </table> I. Find corresponding physical addresses of generated logical addresses. If the physical address is not valid then write “invalid”. II. Map user’s view of the main memory.	Logical Memory		0	110	1	111	2	220	3	221	4	230	5	260	PMT		0	6	1	3	2	0	3	8	4	7	5	12	Main Memory		[6+2]
Logical Memory																																
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	b) Dipu, an operating system developer, is working on memory management. His task is to implement efficient paging on the system he is working on. But recently, due to not increasing his salary, he planned to sabotage the company. He first implemented LRU algorithm and then attempted to implement a new algorithm which he named sub-optimal algorithm, where the page that is demanded in the nearest future is replaced by the new page. The main memory of his system has 3 frames and the page reference requests are as follows: 7, 4, 6, 7, 2, 0, 5, 1, 4, 7, 1, 2 Apply both the above algorithms and track the page faults. Calculate the page fault ratio. Compare between these two algorithms and mention which algorithm performs better.	[5+5 +2]
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